

Ricardo Jaimes Chavez | Tripl3tr3s

AI Engineer | Production LLM & Agentic Systems on Structured Financial Data

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SUMMARY

AI engineer building production agentic systems in high-stakes financial domains. In crypto and decentralized systems since 2020; since 2025 in Mexican fiscal compliance, where a wrong output is a legal filing with the tax authority. Shipped solo: a production MCP server (46 tools), a three-tier Claude orchestration layer with cost-aware model routing, and a full AI-native compliance platform with Chat interface, STT input, prompt engineering, eval coverage, end-to-end observability, human-in-the-loop approval on irreversible operations, and 6,638+ tests at 91% coverage. Previously led development at RetailDAO, where I ran smart-contract audits and executed the on-chain governance action that permanently capped the token supply, removing inflation and honeypot risk for holders. Python and TypeScript/Node. Async collaboration with North American teams; documents English-first.

EXPERIENCE

AI Engineer & Founder, [DISAI](#) (independent)

Jun 2025 – Present

- **Production MCP server (46 tools):** Designed, built and operate a Model Context Protocol server (TypeScript/Node, Railway) covering the full CFDI 4.0 lifecycle: invoicing, payroll, carta porte, payment complements, live SAT catalog injection, EFOS blacklist sync, PAC stamping, multi-tenant credential handling. ~1.2ms median tool dispatch under sustained load.
- **Three-tier agentic orchestration on Claude:** Routing agent classifying 10 fiscal domains, 10 domain agents using native tool use with agentic loops and self-correction, scoped tool allowlists, and cost-aware model routing across Haiku/Sonnet/Opus tiers.
- **Reliability under irreversible operations:** Human-in-the-loop approval gate before any tax-authority stamping, two-layer idempotency, single-flight deduplication on lazy loading, fail-open vs fail-closed policy per subsystem, and circuit breakers on third-party stamping (PAC) providers.
- **Retrieval and context injection:** Domain-scoped retrieval over 54+ MCP resources (SAT catalogs, document templates, regulatory docs, Blacklists) with live per-domain context injection, lazy loading and single-flight deduplication. SAT catalog lookup via fuzzy string matching across 52k+ catalog entries. Chose deterministic resource retrieval over vector search: *fiscal answers must be complete and reproducible, and re-indexing must not silently change last quarter's output.*
- **Evals & Observability:** Versioned prompt templates and system instructions across all agents; eval coverage plus end-to-end tracing (Langfuse) with per-session cost attribution and failure triage. Five-tier test strategy from unit through contract to E2E smoke, 6,638+ tests at 91% coverage.
- **Full-stack:** Next.js 16 / React 19 streaming frontend, Node backend, PostgreSQL/SQLite data layer, Zod schema validation at every boundary, CI/CD.
- **Applied mathematics in production:** Directed supplier-network risk graph with BFS propagation and DFS cycle detection for tax-fraud carousel patterns; derived and published the decay bound, released as the open-source `efos-risk-graph` npm package.
- **Customer-facing delivery.** Ran discovery sessions with Mexican accounting firms and translated ambiguous fiscal and regulatory requirements into working agent workflows.
- **AI-assisted engineering.** Built the platform solo directing Claude Code, Gemini CLI, OpenCode, and other harnesses daily, reviewing all generated code for logic, security and performance defects as the last line of defense before production.

Core Contributor / Lead Developer, [RetailDAO](#) (Base mainnet)

Jun 2024 - Jan 2026

- **Protocol security, audit and irreversible mitigation.** After external scanners flagged \$RETAIL for an active mint function and opaque ownership, I led the audit that traced the root cause to the Aragon `IVotes` plugin retaining `execute()` permissions on the original mint DAO, compounded by non-standard ownership (no `renounceOwnership` path) and proxy contracts obscuring control. Designed and executed the mitigation: permanently revoke all `execute()` permissions, rendering the mint DAO immutable and capping supply at 1B tokens. Passed by on-chain token-weighted vote; external risk flags cleared post-execution. [Proposal on Base mainnet](#) · [Audit and mitigation report](#)
- **Real-time analytics terminal.** Built and shipped a Bloomberg-style crypto terminal aggregating 8+ sources (Federal

Reserve, on-chain, exchanges, intelligence firms) over dual WebSockets, with Redis caching, rate limiting, exponential backoff with jitter, golden-dataset fallback and graceful degradation. React, Node.js, Redis, Docker.

- **Protocol documentation platform.** Authored and shipped the complete documentation site solo: tokenomics, audits, governance structure, treasury and budget policy, onboarding. Docusaurus, deployed on GitHub Pages, plus the GitHub org infrastructure behind it. retaildao.github.io/Docs

Data & Research Analyst, [Coinsider](#)

Mar 2024 - Sep 2025

- Hired after a multi-stage technical assessment on protocol mechanics, market structure and on-chain analysis.
- Delivered 25+ peer-reviewed analytics reports (1,200+ pages): investor-risk models, SWOT analysis, macro forecasting. Translated dense quantitative material into clear writing that reached 70k+ views.

Data Analysis Consultant (short engagement), [Chainlabs](#)

Jun 2025

- Delivered two wallet-analysis notebooks in Python for a European crypto-intelligence firm: OSINT techniques, cross-chain data wrangling, dataset querying, visualizations and relational graphs in Plotly and Matplotlib.

Foundation

2020 – 2024

Self-directed work in crypto and decentralized systems alongside a design career: protocol mechanics (PoW/PoS, L2s, Zero-Knowledge proofs, cross-chain), on-chain operations, multi-wallet and cold-storage management, and market structure and quantitative analysis (derivatives, leverage, risk-adjusted return metrics, systematic strategies). Externally validated by passing Coinsider's technical assessment in 2024.

Prior to that: 12+ years as a graphic designer, illustrator and tattoo artist/studio owner, before transitioning full-time into software and AI engineering.

SKILLS

AI / LLM: LLM application development, AI agents, multi-agent orchestration, agentic loops and self-correction, prompt engineering, system instructions, evals, hallucination control, human-in-the-loop (HITL), cost-aware model routing, retrieval and context injection over MCP resources, fuzzy/lexical matching, Claude API, tool use / function calling, MCP (full primitive set: tools, resources, prompts, elicitation, MCP UI apps)

Languages & Backend: Python, TypeScript, Node.js, SQL, REST APIs, object-oriented design (abstract base classes, interface-based DI, strategy pattern), multi-service architecture, streaming / server-sent events (SSE), Streamable HTTP.

Data & Infrastructure: PostgreSQL, SQLite, Redis, Docker, WebSockets, rate limiting and backoff strategies, caching, circuit breakers, idempotency.

Frontend: React, Next.js, Tailwind CSS, shadcn/ui

DevOps & Practices: Docker, CI/CD, Git, Gitea, Coolify, Railway, automated testing (Vitest), schema validation (Zod), observability (Langfuse), security, permissions, auditability, technical documentation

AI-assisted engineering: Claude Code, Gemini CLI, Zed, OpenCode (daily production use), AI-driven code review

Customer-facing: Customer discovery, requirements gathering, solution delivery, UAT, stakeholder collaboration

SELECTED PROJECTS

- **SAT-MCP Server:** Production MCP server for Mexican fiscal compliance. 46 tools, 18 prompts, 54+ resources, full MCP primitive set plus in-chat mini-apps; SSE and streamable HTTP transport; 6,638+ tests, 91%+ coverage.
- **DISAI-Conta:** AI-native fiscal compliance platform for accounting firms. Routing agent plus 10 domain agents, scoped allowlists, live context injection, full observability, human-in-the-loop compliance dashboard.
- **Open-source pattern libraries** (reusable production patterns extracted from projects): **efos-risk-graph** (graph risk propagation, BFS/DFS), **agentic-tool-loop**, **llm-cost-router**, **mcp-tool-idempotency** and **euler-worked-out**.
- **n8n AI Automation Stack:** Self-hosted n8n ultra-lean instance with direct MCP server/client integration and sub-agents per document type; replaced Zapier/Make for ~80% recurring-cost reduction.

EDUCATION & CERTIFICATIONS

- MIT OCW 6.1200J, Mathematics for CS. Self-study, in progress.
- Model Context Protocol: Advanced Topics, Anthropic (2025)
- Claude Code in Action, Anthropic (2025)

- Building Scalable Agentic Systems, DataCamp (2025)
- n8n Self-Hosted for Enterprises, n8n (2025)
- B.A. Graphic Design, Universidad Autonoma de Aguascalientes (2013)

LANGUAGES

Spanish: Native | English: Professional working proficiency (technical documentation, proposals, async collaboration with US-based teams)